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(54) MATCHING NETWORK FOR A POWER AMPLIFIER

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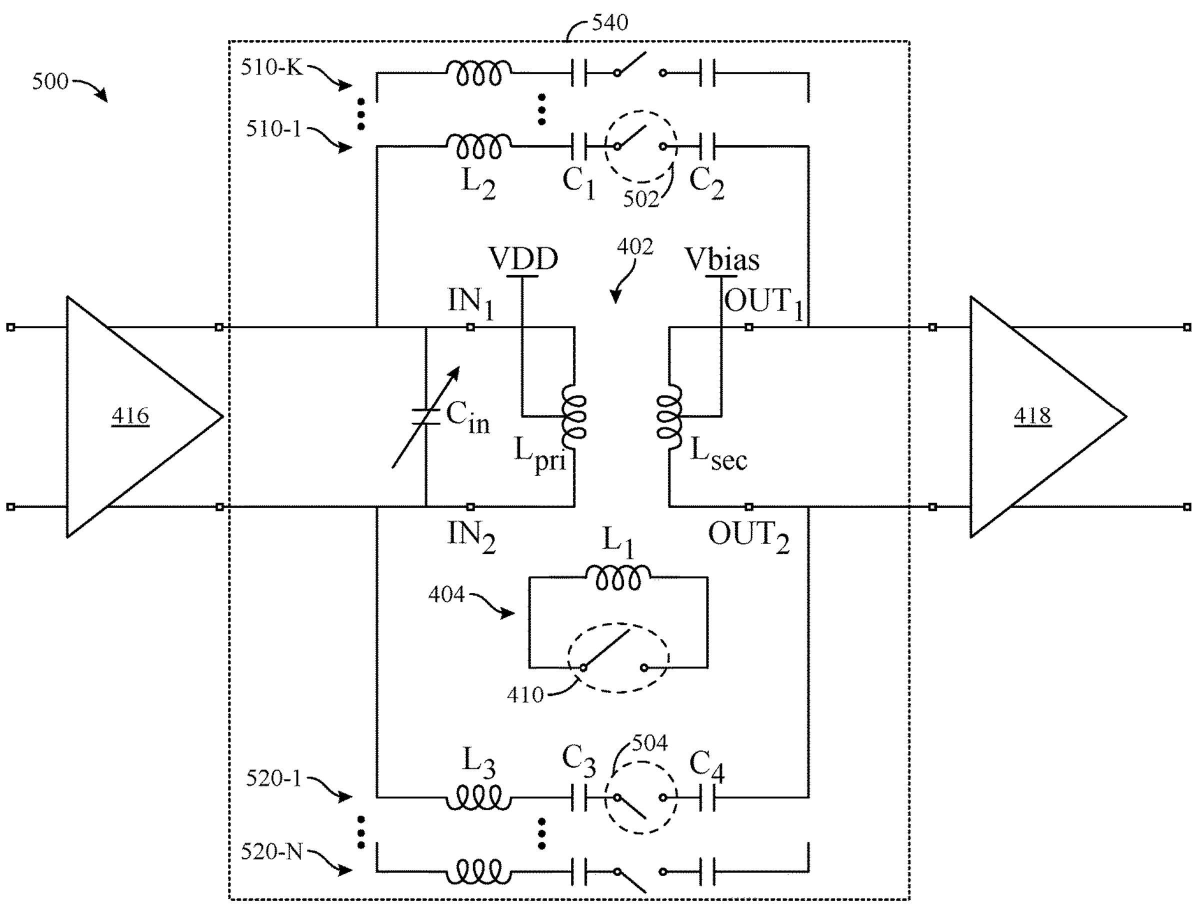
(57) **ABSTRACT**

Methods and apparatus for transferring a signal from a first stage of an amplification circuit to a second stage of an amplification circuit via a matching network are described. An example matching network generally includes a transformer, a first coil selectively coupled, inductively, to the transformer, and inductive path(s) coupled between a first input node and a first output node of the transformer. Each inductive path includes a respective second coil selectively coupled, inductively, to the transformer. The transformer includes: the first input node and a second input node for coupling to the first stage; the first output node and a second output node for coupling to the second stage; a primary winding coupled between the first input node and the second input node; and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node.



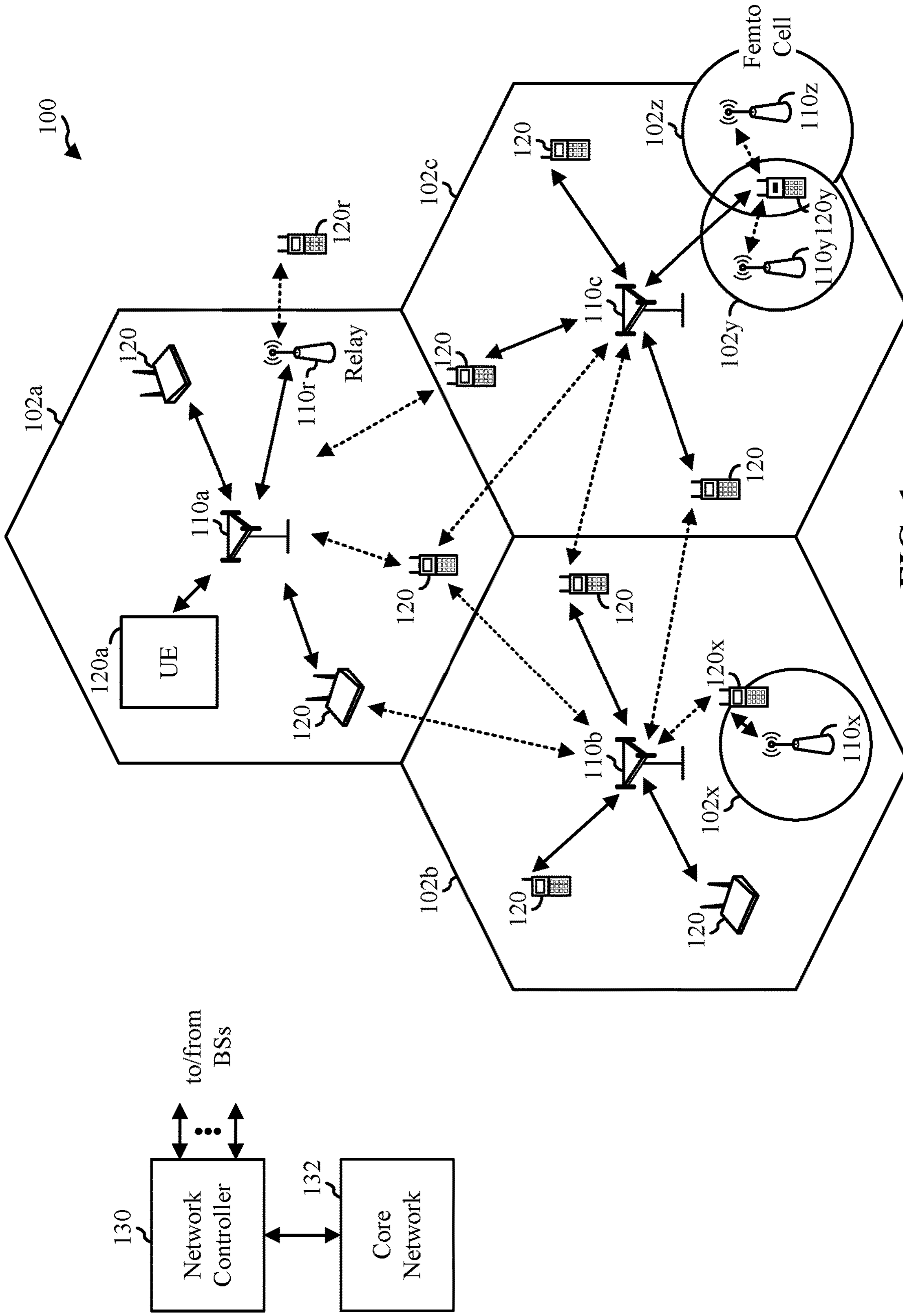
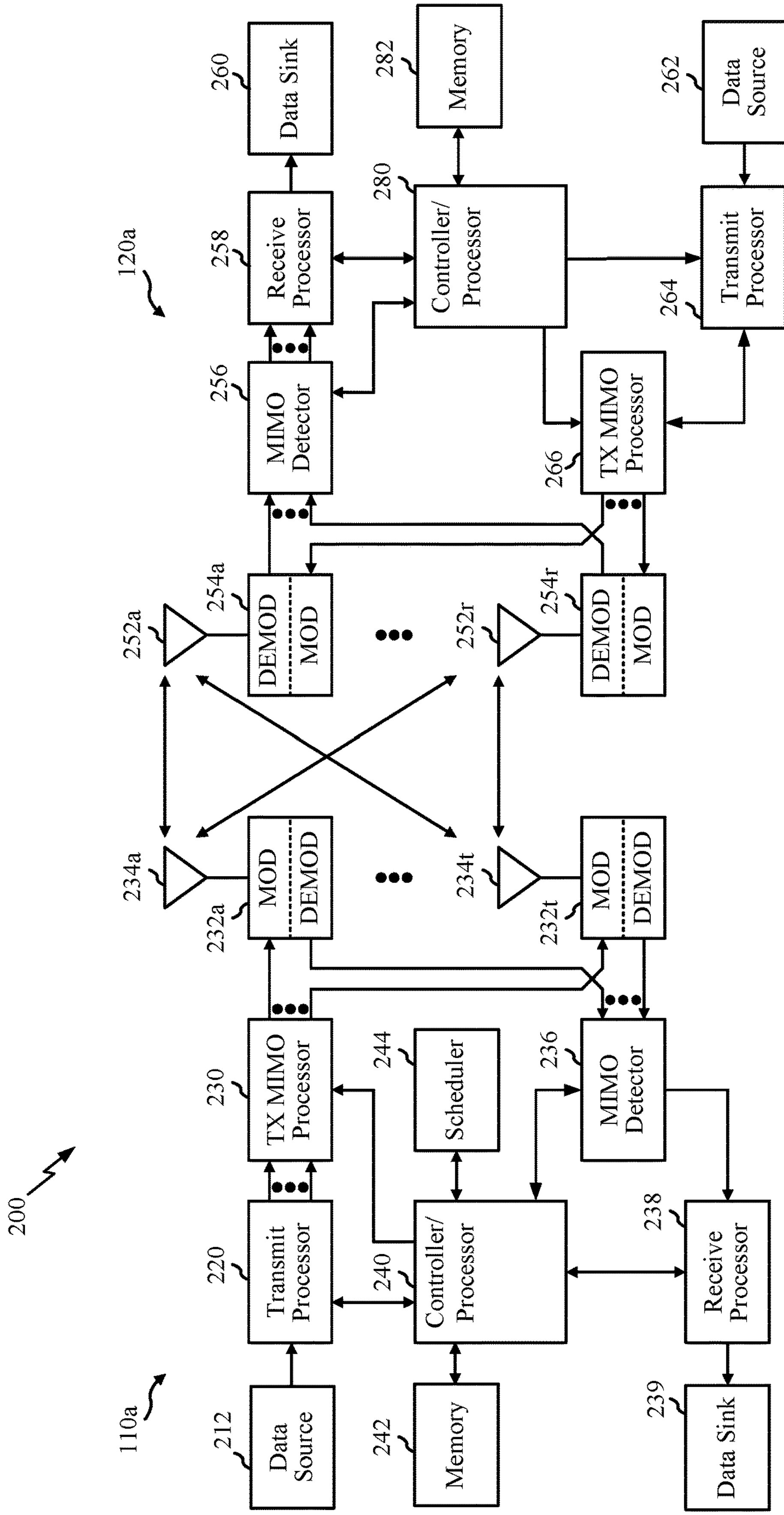

100
102a
102b
102c
102x
102y
102z
110a
110b
110c
110r
110x
110y
110z
120
120a
120r
120x
120y
UE
Relay
Femto Cell
130
Network Controller
132
Core Network
to/from BSs

FIG. 1



200
110a
120a
212
Data Source
220
Transmit Processor
230
TX MIMO Processor
232a
MOD
DEMOD
234a
232t
234t
240
Controller/ Processor
242
Memory
244
Scheduler
236
MIMO Detector
238
Receive Processor
239
Data Sink
252a
254a
252r
254r
256
MIMO Detector
258
Receive Processor
260
Data Sink
280
Controller/ Processor
282
Memory
262
Data Source
264
Transmit Processor
266
TX MIMO Processor

FIG. 2

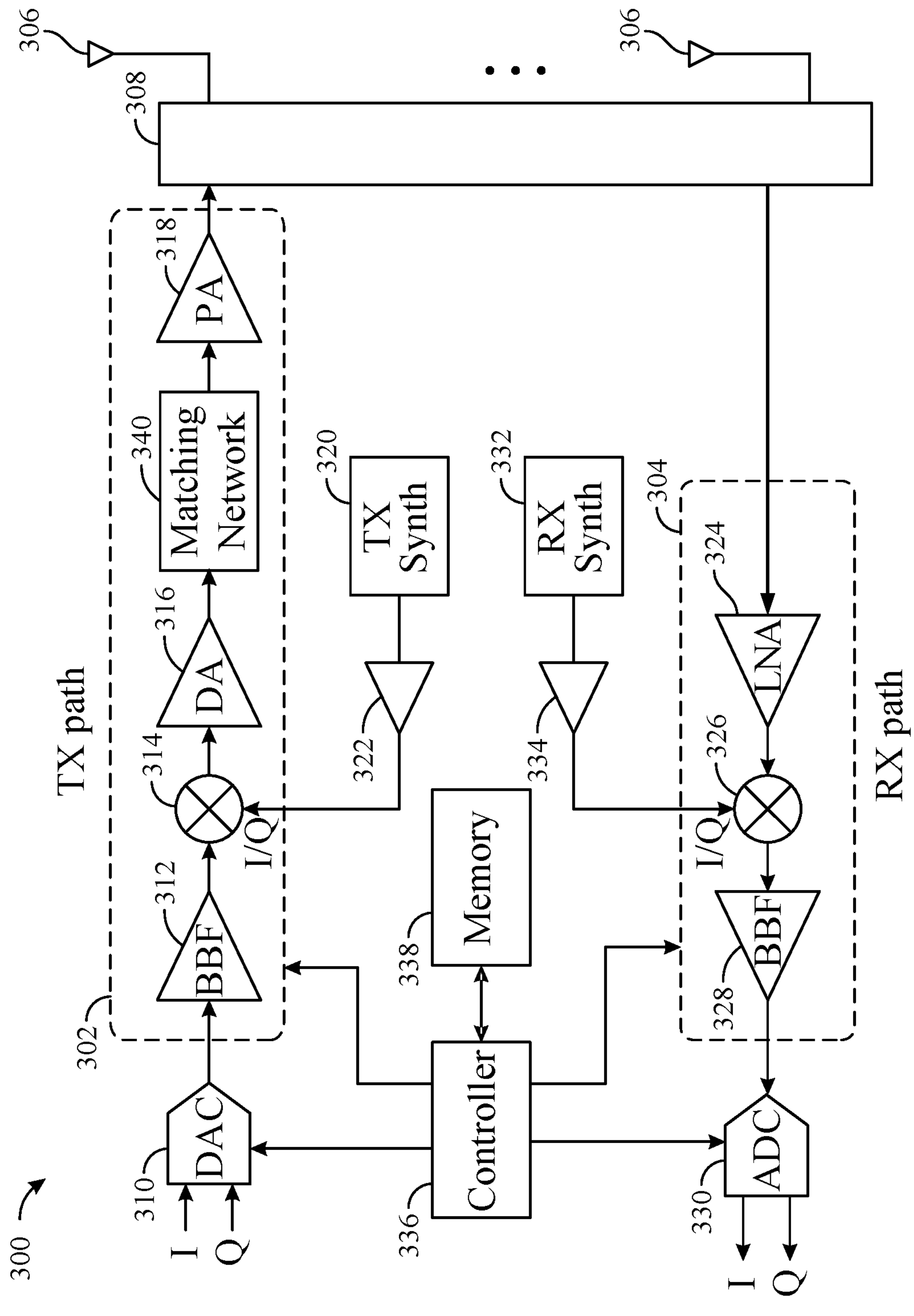

300
306
306
308
TX path
302
310
DAC
I
Q
312
BBF
314
I/Q
316
DA
340
Matching Network
318
PA
320
TX Synth
322
332
RX Synth
334
Memory
338
Controller
336
304
RX path
324
LNA
326
I/Q
328
BBF
330
ADC
I
Q

FIG. 3

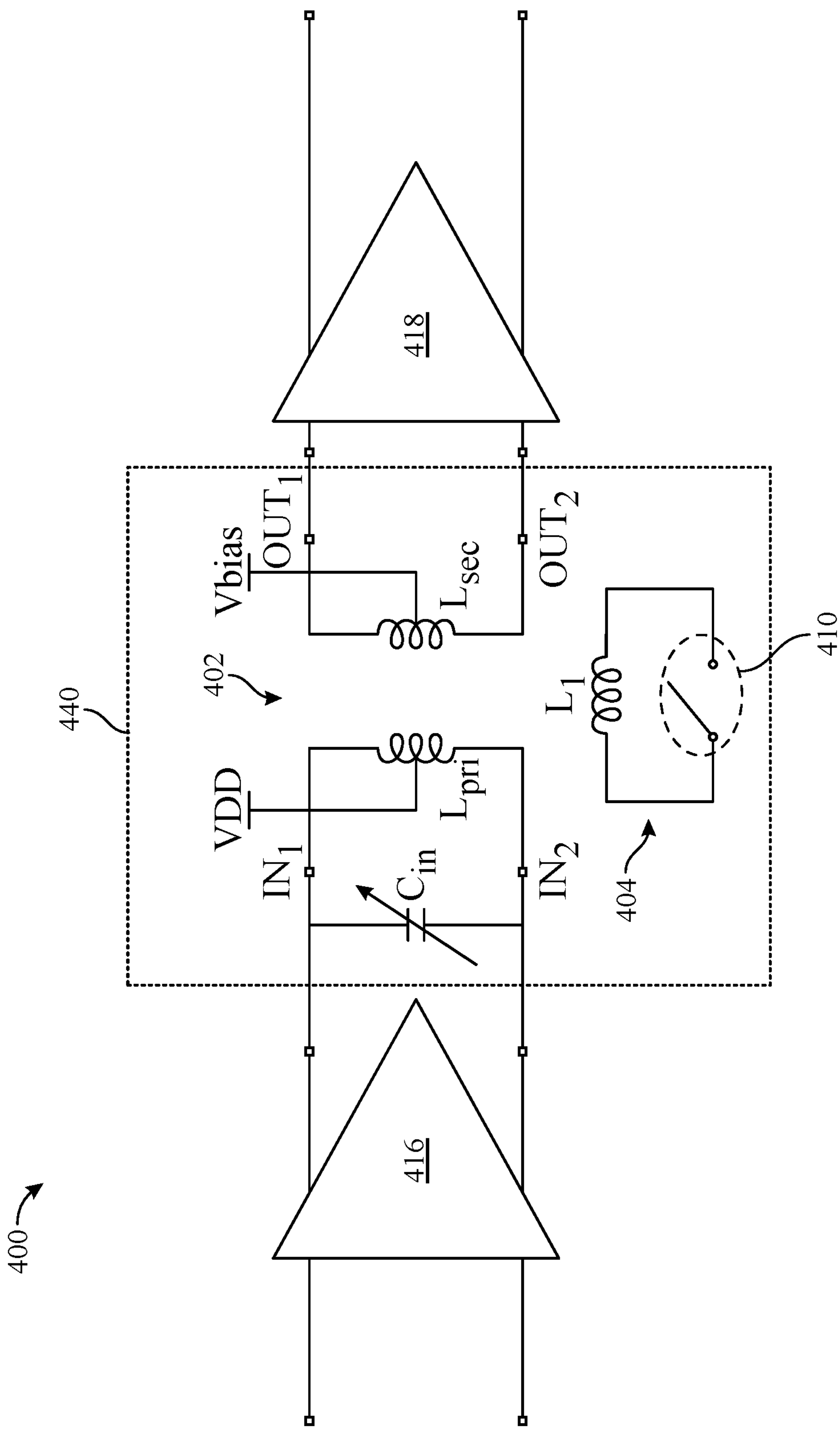

400
416
418
440
402
404
410
VDD
Vbias
IN1
IN2
OUT1
OUT2
Lpri
Lsec
L1
Cin

FIG. 4

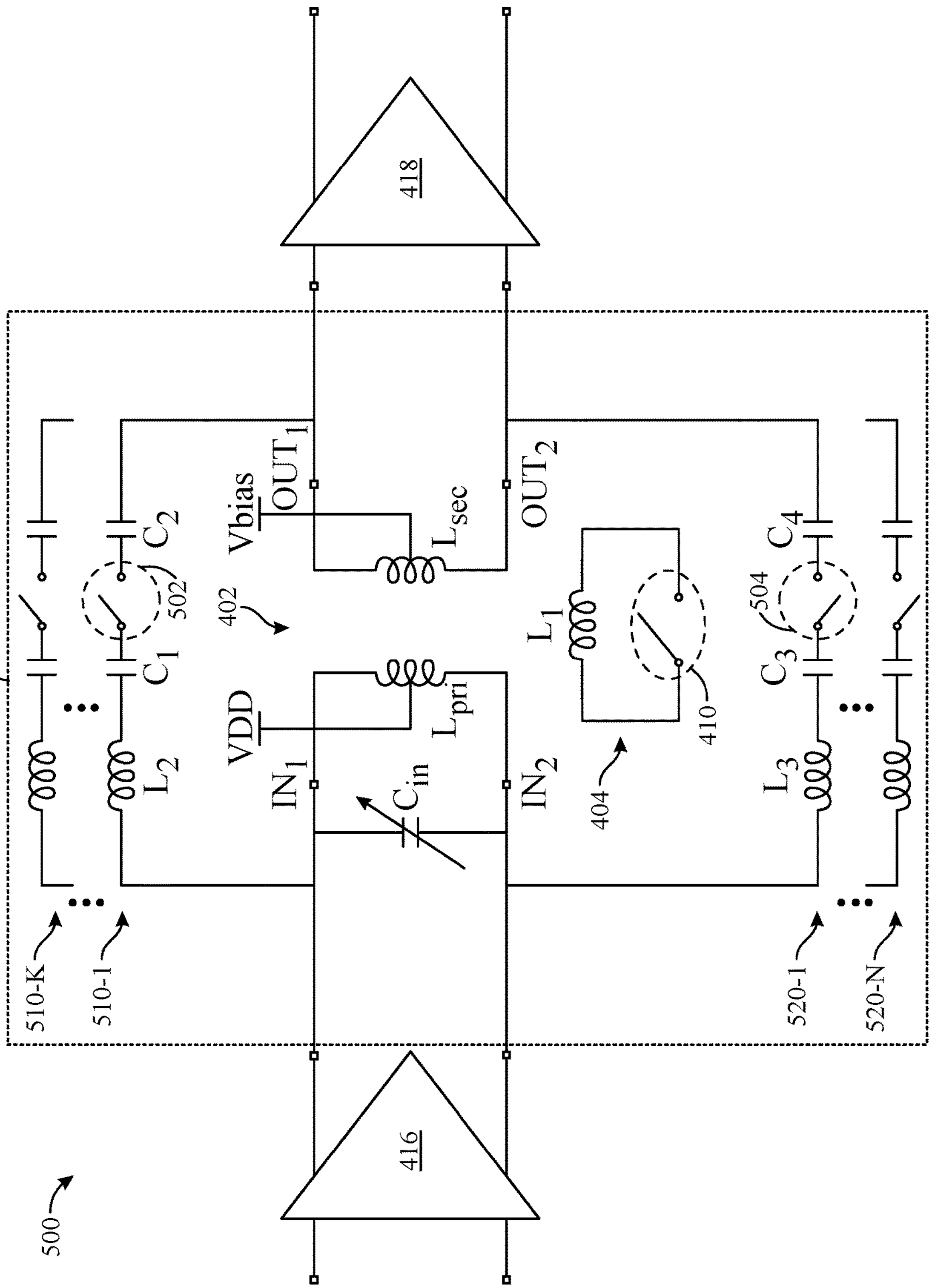

500
416
418
540
510-K
510-1
520-1
520-N
IN1
IN2
OUT1
OUT2
VDD
Vbias
Lpri
Lsec
Cin
L1
L2
L3
C1
C2
C3
C4
402
404
410
502
504

FIG. 5

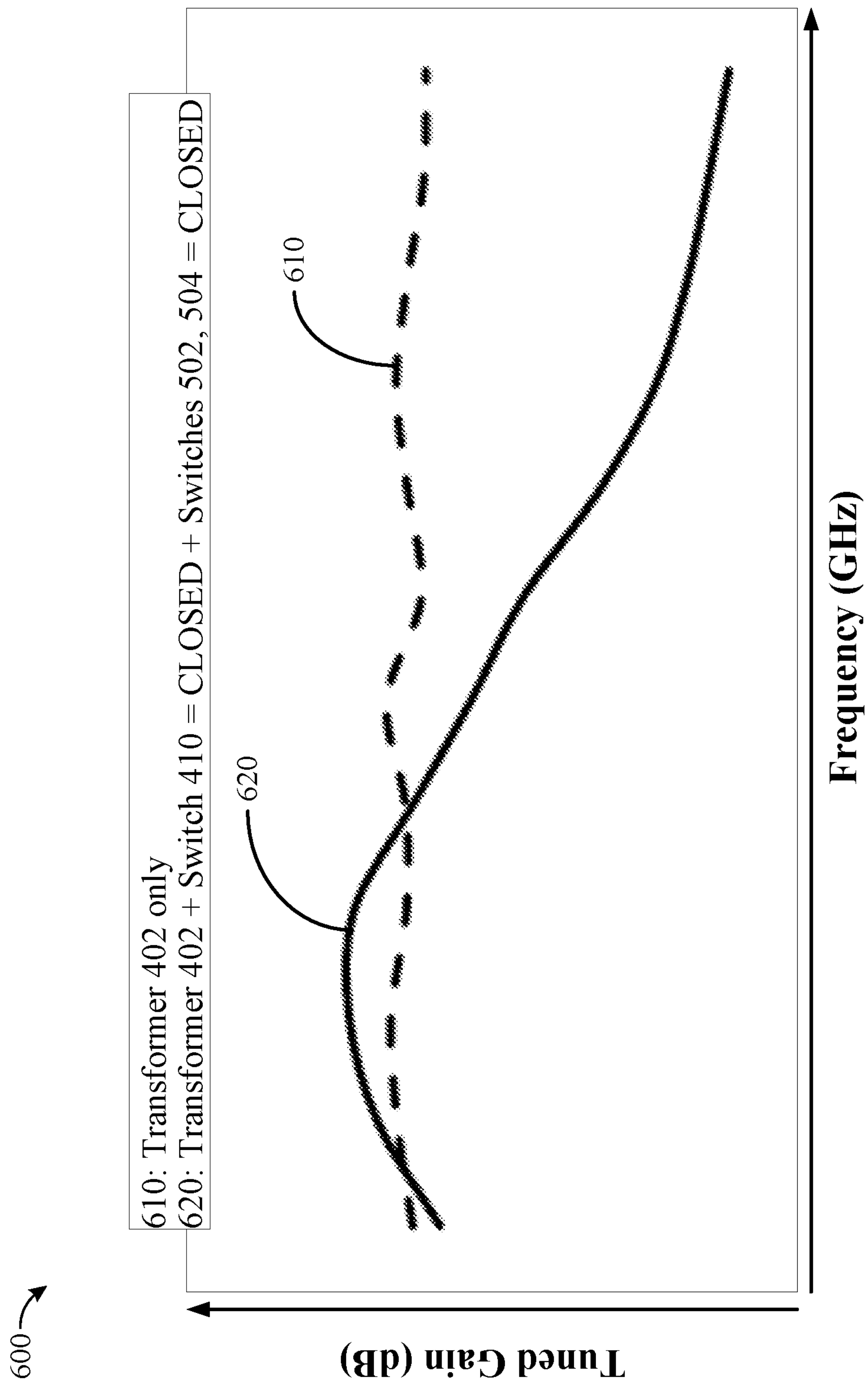

600
610: Transformer 402 only
620: Transformer 402 + Switch 410 = CLOSED + Switches 502, 504 = CLOSED
610
620
Tuned Gain (dB)
Frequency (GHz)

FIG. 6

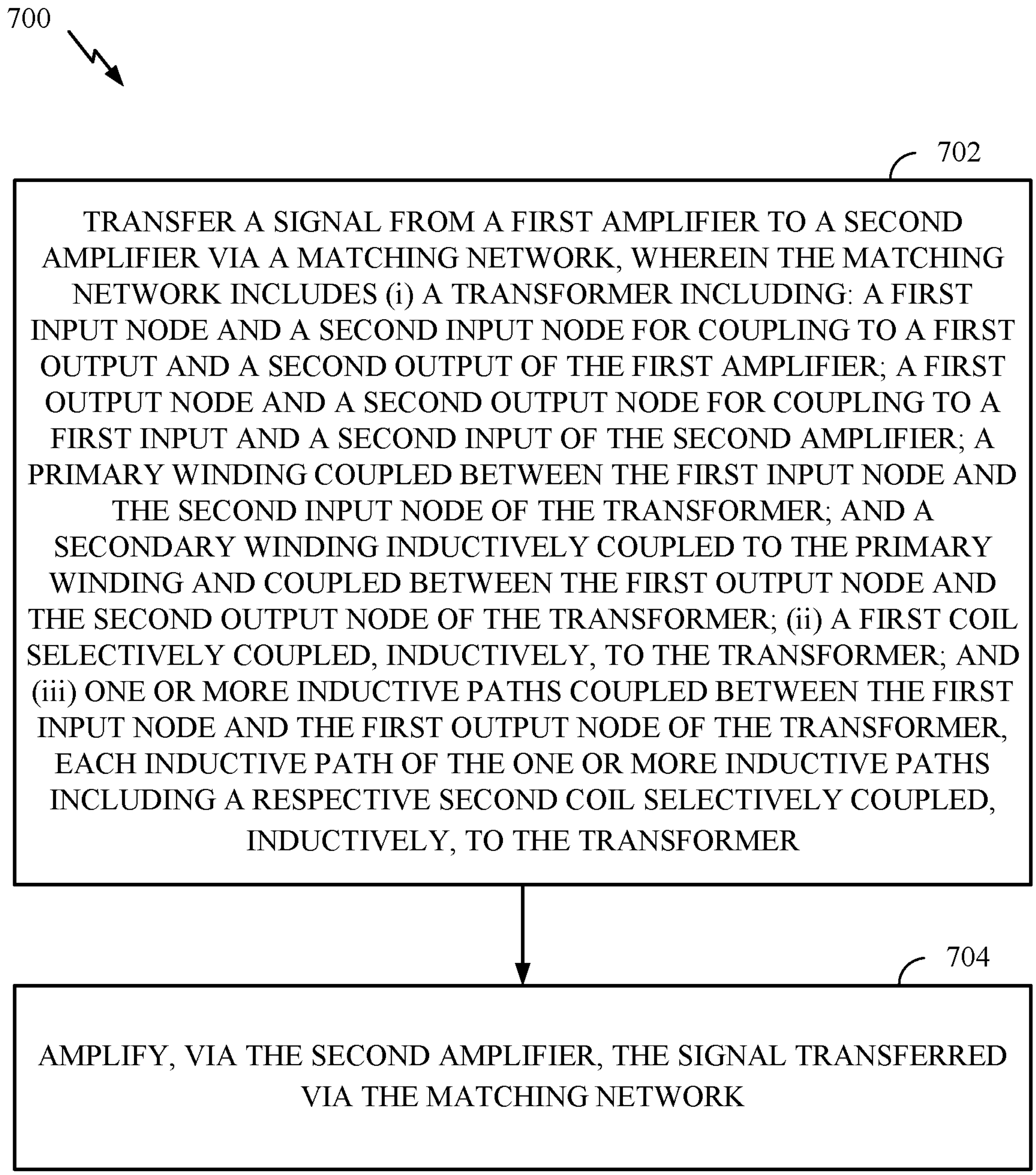

700
702
TRANSFER A SIGNAL FROM A FIRST AMPLIFIER TO A SECOND AMPLIFIER VIA A MATCHING NETWORK, WHEREIN THE MATCHING NETWORK INCLUDES (i) A TRANSFORMER INCLUDING: A FIRST INPUT NODE AND A SECOND INPUT NODE FOR COUPLING TO A FIRST OUTPUT AND A SECOND OUTPUT OF THE FIRST AMPLIFIER; A FIRST OUTPUT NODE AND A SECOND OUTPUT NODE FOR COUPLING TO A FIRST INPUT AND A SECOND INPUT OF THE SECOND AMPLIFIER; A PRIMARY WINDING COUPLED BETWEEN THE FIRST INPUT NODE AND THE SECOND INPUT NODE OF THE TRANSFORMER; AND A SECONDARY WINDING INDUCTIVELY COUPLED TO THE PRIMARY WINDING AND COUPLED BETWEEN THE FIRST OUTPUT NODE AND THE SECOND OUTPUT NODE OF THE TRANSFORMER; (ii) A FIRST COIL SELECTIVELY COUPLED, INDUCTIVELY, TO THE TRANSFORMER; AND (iii) ONE OR MORE INDUCTIVE PATHS COUPLED BETWEEN THE FIRST INPUT NODE AND THE FIRST OUTPUT NODE OF THE TRANSFORMER, EACH INDUCTIVE PATH OF THE ONE OR MORE INDUCTIVE PATHS INCLUDING A RESPECTIVE SECOND COIL SELECTIVELY COUPLED, INDUCTIVELY, TO THE TRANSFORMER
704
AMPLIFY, VIA THE SECOND AMPLIFIER, THE SIGNAL TRANSFERRED VIA THE MATCHING NETWORK

FIG. 7

MATCHING NETWORK FOR A POWER AMPLIFIER

TECHNICAL FIELD

[0001] Certain aspects of the present disclosure generally relate to electronic circuits and, more particularly, to a matching network, such as an impedance matching network for a power amplifier for use in a transmitter architecture.

BACKGROUND

[0002] Wireless communication devices are widely deployed to provide various communication services such as telephony, video, data, messaging, broadcasts, and so on. Such wireless communication devices may transmit and/or receive radio frequency (RF) signals via any of various suitable radio access technologies (RATs) including, but not limited to, Fifth Generation (5G) New Radio (NR), Long Term Evolution (LTE), Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Wideband CDMA (WCDMA), Global System for Mobility (GSM), Bluetooth, Bluetooth Low Energy (BLE), ZigBee, wireless local area network (WLAN) RATs (e.g., WiFi), and the like.

[0003] A wireless communication network may include a number of base stations that can support communication for a number of mobile stations. A mobile station (MS) may communicate with a base station (BS) via a downlink and an uplink. The downlink (or forward link) refers to the communication link from the base station to the mobile station, and the uplink (or reverse link) refers to the communication link from the mobile station to the base station. A base station may transmit data and control information on the downlink to a mobile station and/or may receive data and control information on the uplink from the mobile station. The base station and/or mobile station may include radio frequency front-end (RFFE) circuitry for processing and amplifying signals for transmission and reception. For example, the RFFE circuitry may include a power amplifier (PA) for amplifying a radio frequency signal for transmission. Moreover, a driver amplifier (DA) (also referred to as a "pre-PA") may be used to generate signals to drive an input of the PA.

SUMMARY

[0004] The systems, methods, and devices of the disclosure each have several aspects, no single one of which is solely responsible for its desirable attributes. Without limiting the scope of this disclosure as expressed by the claims which follow, some features will now be discussed briefly. After considering this discussion, and particularly after reading the section entitled "Detailed Description," one will understand how the features of this disclosure provide advantages that include higher gain bandwidths, higher saturated output power, flatter gain response across a target frequency range, and improved quality (Q) factor of a power amplifier (PA).

[0005] Certain aspects of the present disclosure provide a matching network for an amplification circuit. The matching network includes a transformer. The transformer includes a first input node and a second input node for coupling to a first stage of the amplification circuit, a first output node and a second output node for coupling to a second stage of the amplification circuit, a primary winding coupled between the first input node and the second input node of the transformer, and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer. The matching network also includes a first coil selectively coupled, inductively, to the transformer. The matching network further includes one or more first inductive paths coupled between the first input node and the first output node of the transformer. Each inductive path of the one or more first inductive paths includes a respective second coil selectively coupled, inductively, to the transformer.

[0006] Certain aspects of the present disclosure provide a wireless device. The wireless device includes a first amplifier, a second amplifier, and a matching network coupled between the first amplifier and the second amplifier. The matching network includes a transformer. The transformer includes a first input node and a second input node for coupling to a first output and a second output of the first amplifier, a first output node and a second output node for coupling to a first input and a second input of the second amplifier, a primary winding coupled between the first input node and the second input node of the transformer; and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer. The matching network also includes a first coil selectively coupled, inductively, to the transformer. The matching network further includes one or more first inductive paths coupled between the first input node and the first output node of the transformer. Each inductive path of the one or more first inductive paths includes a respective second coil selectively coupled, inductively, to the transformer.

[0007] Certain aspects of the present disclosure provide a method of wireless communication. The method generally includes transferring a signal from a first amplifier to a second amplifier via a matching network. The matching network includes a transformer. The transformer includes a first input node and a second input node for coupling to a first output and a second output of the first amplifier, a first output node and a second output node for coupling to a first input and a second input of the second amplifier, a primary winding coupled between the first input node and the second input node of the transformer, and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer. The transformer also includes a first coil selectively coupled, inductively, to the transformer. The transformer further includes one or more inductive paths coupled between the first input node and the first output node of the transformer. Each inductive path of the one or more inductive paths includes a respective second coil selectively coupled, inductively, to the transformer. The method also includes amplifying, via the second amplifier, the signal transferred via the matching network.

[0008] To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects.

[0010] FIG. **1** is a diagram of an example wireless communications network, in which aspects of the present disclosure may be practiced.

[0011] FIG. **2** is a block diagram conceptually illustrating a design of an example a base station (BS) and user equipment (UE), in which aspects of the present disclosure may be practiced.

[0012] FIG. **3** is a block diagram of an example radio frequency (RF) transceiver, in which aspects of the present disclosure may be practiced.

[0013] FIG. **4** illustrates an example amplifier circuit, in accordance with certain aspects of the present disclosure.

[0014] FIG. **5** illustrates another example amplifier circuit, in accordance with certain aspects of the present disclosure.

[0015] FIG. **6** illustrates a graph of different final tuned gains of an amplifier circuit, in accordance with certain aspects of the present disclosure.

[0016] FIG. **7** is a flow diagram of example operations for amplifying a signal for transmission, in accordance with certain aspects of the present disclosure.

[0017] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one aspect may be beneficially utilized on other aspects without specific recitation.

DETAILED DESCRIPTION

[0018] Certain aspects of the present disclosure generally relate to electronic components and, more particularly, to an inter-stage matching network for an amplifier (e.g., power amplifier or "PA"). For example, the matching network described herein may be coupled between a first stage of an amplifier circuit and a second stage of the amplifier circuit. The first stage may include a first amplifier (e.g., a pre-power amplifier or "pre-PA"), and the second stage may include a second amplifier (e.g., a power amplifier or "PA"). As described in greater detail herein, the matching network may include, without limitation, a transformer, a first coil selectively coupled, inductively, to the transformer, and one or more inductive paths coupled between an input node and an output node of the transformer. Each inductive path may include a respective second coil that is selectively coupled, inductively, to the transformer.

[0019] The matching network may be used for impedance matching between stages of the multi-stage amplifier circuit. The matching network may allow for improved performance of an amplifier (e.g., PA) in terms of higher gain bandwidths, higher saturated output power (Psat), flatter gain response across the target frequency range, and improved (e.g., higher) quality factor (also referred to as "Q factor" or "Q"), as illustrative, non-limiting examples.

[0020] Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Based on the teachings herein one skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0021] The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects.

[0022] As used herein, the term "connected with" in the various tenses of the verb "connect" may mean that element A is directly connected to element B or that other elements may be connected between elements A and B (i.e., that element A is indirectly connected with element B). In the case of electrical components, the term "connected with" may also be used herein to mean that a wire, trace, or other electrically conductive material is used to electrically connect elements A and B (and any components electrically connected therebetween).

An Example Wireless System

[0023] FIG. **1** illustrates an example wireless communications network **100**, in which aspects of the present disclosure may be practiced. For example, the wireless communications network **100** may be a New Radio (NR) system (e.g., a Fifth Generation (5G) NR network), a Sixth Generation (6G) cellular system, an Evolved Universal Terrestrial Radio Access (E-UTRA) system (e.g., a Fourth Generation (4G) network), a Universal Mobile Telecommunications System (UMTS) (e.g., a Second Generation/Third Generation (2G/3G) network), or a code division multiple access (CDMA) system (e.g., a 2G/3G network), or may be configured for communications according to an Institute of Electrical and Electronics Engineers (IEEE) standard such as one or more of the 802.11 standards, etc.

[0024] As illustrated in FIG. **1**, the wireless communications network **100** may include a number of base stations (BSs) **110***a-z* (each also individually referred to herein as "BS **110**" or collectively as "BSs **110**") and other network entities. A BS may also be referred to as an access point (AP), an evolved Node B (eNodeB or eNB), a next generation Node B (gNodeB or gNB), or some other terminology.

[0025] A BS **110** may provide communication coverage for a particular geographic area, sometimes referred to as a "cell," which may be stationary or may move according to the location of a mobile BS. In some examples, the BSs **110**

may be interconnected to one another and/or to one or more other BSs or network nodes (not shown) in wireless communications network **100** through various types of backhaul interfaces (e.g., a direct physical connection, a wireless connection, a virtual network, or the like) using any suitable transport network. In the example shown in FIG. **1**, the BSs **110***a*, **110***b*, and **110***c* may be macro BSs for the macro cells **102***a*, **102***b*, and **102***c*, respectively. The BS **110***x* may be a pico BS for a pico cell **102***x*. The BSs **110***y* and **110***z* may be femto BSs for the femto cells **102***y* and **102***z*, respectively. A BS may support one or multiple cells.

[0026] The BSs **110** communicate with one or more user equipments (UEs) **120***a-y* (each also individually referred to herein as "UE **120**" or collectively as "UEs **120**") in the wireless communications network **100**. A UE may be fixed or mobile and may also be referred to as a user terminal (UT), a mobile station (MS), an access terminal, a station (STA), a client, a wireless device, a mobile device, or some other terminology. A user terminal may be a wireless device, such as a cellular phone, a smartphone, a personal digital assistant (PDA), a handheld device, a wearable device, a wireless modem, a laptop computer, a tablet, a personal computer, etc.

[0027] The BSs **110** are considered transmitting entities for the downlink and receiving entities for the uplink. The UEs **120** are considered transmitting entities for the uplink and receiving entities for the downlink. As used herein, a "transmitting entity" is an independently operated apparatus or device capable of transmitting data via a frequency channel, and a "receiving entity" is an independently operated apparatus or device capable of receiving data via a frequency channel. In the following description, the subscript "dn" denotes the downlink, the subscript "up" denotes the uplink. N_{up} UEs may be selected for simultaneous transmission on the uplink, N_{dn} UEs may be selected for simultaneous transmission on the downlink. N_{up} may or may not be equal to N_{dn}, and N_{up} and N_{dn} may be static values or can change for each scheduling interval. Beam-steering or some other spatial processing technique may be used at the BSs **110** and/or UEs **120**.

[0028] The UEs **120** (e.g., **120***x*, **120***y*, etc.) may be dispersed throughout the wireless communications network **100**, and each UE **120** may be stationary or mobile. The wireless communications network **100** may also include relay stations (e.g., relay station **110***r*), also referred to as relays or the like, that receive a transmission of data and/or other information from an upstream station (e.g., a BS **110***a* or a UE **120***r*) and send a transmission of the data and/or other information to a downstream station (e.g., a UE **120** or a BS **110**), or that relays transmissions between UEs **120**, to facilitate communication between devices.

[0029] The BSs **110** may communicate with one or more UEs **120** at any given moment on the downlink and uplink. The downlink (i.e., forward link) is the communication link from the BSs **110** to the UEs **120**, and the uplink (i.e., reverse link) is the communication link from the UEs **120** to the BSs **110**. A UE **120** may also communicate peer-to-peer with another UE **120**.

[0030] The wireless communications network **100** may use multiple transmit and multiple receive antennas for data transmission on the downlink and uplink. BSs **110** may be equipped with a number Nap of antennas to achieve transmit diversity for downlink transmissions and/or receive diversity for uplink transmissions. A set Nu of UEs **120** may receive downlink transmissions and transmit uplink transmissions. Each UE **120** may transmit user-specific data to and/or receive user-specific data from the BSs **110**. In general, each UE **120** may be equipped with one or multiple antennas. The Nu UEs **120** can have the same or different numbers of antennas.

[0031] The wireless communications network **100** may be a time division duplex (TDD) system or a frequency division duplex (FDD) system. For a TDD system, the downlink and uplink share the same frequency band. For an FDD system, the downlink and uplink use different frequency bands. The wireless communications network **100** may also utilize a single carrier or multiple carriers for transmission. Each UE **120** may be equipped with a single antenna (e.g., to keep costs down) or multiple antennas (e.g., where the additional cost can be supported).

[0032] A network controller **130** (also sometimes referred to as a "system controller") may be in communication with a set of BSs **110** and provide coordination and control for these BSs **110** (e.g., via a backhaul). In certain cases (e.g., in a 5G NR system), the network controller **130** may include a centralized unit (CU) and/or a distributed unit (DU). In certain aspects, the network controller **130** may be in communication with a core network **132** (e.g., a 5G Core Network (5GC)), which provides various network functions such as Access and Mobility Management, Session Management, User Plane Function, Policy Control Function, Authentication Server Function, Unified Data Management, Application Function, Network Exposure Function, Network Repository Function, Network Slice Selection Function, etc.

[0033] In certain aspects of the present disclosure, the BSs **110** and/or the UEs **120** may include a transceiver front end (TX/RX) (also known as a radio frequency front end (RFFE)), which includes a matching network coupled between a first amplifier (e.g., first stage of an amplifier circuit) and a second amplifier (e.g., second stage of the amplifier circuit). As described in more detail herein, the matching network may be implemented with a transformer, a first coil selectively coupled, inductively, to the transformer, and one or more inductive paths coupled between an input node and an output node of the transformer. Each inductive path may include a respective second coil that is selectively coupled, inductively, to the transformer.

[0034] FIG. **2** illustrates example components of BS **110***a* and UE **120***a* (e.g., from the wireless communications network **100** of FIG. **1**), in which aspects of the present disclosure may be implemented.

[0035] On the downlink, at the BS **110***a*, a transmit processor **220** may receive data from a data source **212**, control information from a controller/processor **240**, and/or possibly other data (e.g., from a scheduler **244**). The various types of data may be sent on different transport channels. For example, the control information may be designated for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical hybrid automatic repeat request (HARQ) indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), etc. The data may be designated for the physical downlink shared channel (PDSCH), etc. A medium access control (MAC)-control element (MAC-CE) is a MAC layer communication structure that may be used for control command exchange between wireless nodes. The MAC-CE may be carried in a shared channel such as a

PDSCH, a physical uplink shared channel (PUSCH), or a physical sidelink shared channel (PSSCH).

[0036] The processor **220** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. The transmit processor **220** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH, demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).

[0037] A transmit (TX) multiple-input, multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **232***a*-**232***t*. Each modulator in transceivers **232***a*-**232***t* may process a respective output symbol stream (e.g., for orthogonal frequency division multiplexing (OFDM), etc.) to obtain an output sample stream. Each of the transceivers **232***a*-**232***t* may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the transceivers **232***a*-**232***t* may be transmitted via the antennas **234***a*-**234***t*, respectively.

[0038] At the UE **120***a*, the antennas **252***a*-**252***r* may receive the downlink signals from the BS **110***a* and may provide received signals to the transceivers **254***a*-**254***r*, respectively. The transceivers **254***a*-**254***r* may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each demodulator (DEMOD) in the transceivers **232***a*-**232***t* may further process the input samples (e.g., for OFDM, etc.) to obtain received symbols. A MIMO detector **256** may obtain received symbols from all the demodulators in transceivers **254***a*-**254***r*, perform MIMO detection on the received symbols if applicable, and provide detected symbols. A receive processor **258** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **120***a* to a data sink **260**, and provide decoded control information to a controller/processor **280**.

[0039] On the uplink, at UE **120***a*, a transmit processor **264** may receive and process data (e.g., for the physical uplink shared channel (PUSCH)) from a data source **262** and control information (e.g., for the physical uplink control channel (PUCCH)) from the controller/processor **280**. The transmit processor **264** may also generate reference symbols for a reference signal (e.g., the sounding reference signal (SRS)). The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modulators (MODs) in transceivers **254***a*-**254***r* (e.g., for single-carrier frequency division multiplexing (SC-FDM), etc.), and transmitted to the BS **110***a*. At the BS **110***a*, the uplink signals from the UE **120***a* may be received by the antennas **234**, processed by the demodulators in transceivers **232***a*-**232***t*, detected by a MIMO detector **236** if applicable, and further processed by a receive processor **238** to obtain decoded data and control information sent by the UE **120***a*. The receive processor **238** may provide the decoded data to a data sink **239** and the decoded control information to the controller/processor **240**.

[0040] The memories **242** and **282** may store data and program codes for BS **110***a* and UE **120***a*, respectively. The memories **242** and **282** may also interface with the controllers/processors **240** and **280**, respectively. A scheduler **244** may schedule UEs for data transmission on the downlink and/or uplink.

[0041] Antennas **252**, processors **258**, **264**, **266**, and/or controller/processor **280** of the UE **120***a* and/or antennas **234**, processors **220**, **230**, **238**, and/or controller/processor **240** of the BS **110***a* may be used to perform the various techniques and methods described herein.

[0042] In certain aspects of the present disclosure, the transceivers **232** and/or the transceivers **254** may include a matching network coupled between a first amplifier (e.g., first stage of an amplifier circuit) and a second amplifier (e.g., second stage of the amplifier circuit). As described in more detail herein, the matching network may be implemented with a transformer, a first coil selectively coupled, inductively, to the transformer, and one or more inductive paths coupled between an input node and an output node of the transformer. Each inductive path may include a respective second coil that is selectively coupled, inductively, to the transformer.

[0043] NR may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. NR may support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth into multiple orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers may be dependent on the system bandwidth. The system bandwidth may also be partitioned into subbands. For example, a subband may cover multiple resource blocks (RBs).

Example RF Transceiver

[0044] FIG. **3** is a block diagram of an example radio frequency (RF) transceiver circuit **300**, in accordance with certain aspects of the present disclosure. The RF transceiver circuit **300** includes at least one transmit (TX) path **302** (also known as a "transmit chain") for transmitting signals via one or more antennas **306** and at least one receive (RX) path **304** (also known as a "receive chain") for receiving signals via the antennas **306**. When the TX path **302** and the RX path **304** share an antenna **306**, the paths may be connected with the antenna via an interface **308**, which may include any of various suitable RF devices, such as a switch, a duplexer, a diplexer, a multiplexer, and the like.

[0045] Receiving in-phase (I) and/or quadrature (Q) baseband analog signals from a digital-to-analog converter (DAC) **310**, the TX path **302** may include a baseband filter (BBF) **312**, a mixer **314**, a driver amplifier (DA) **316**, a matching network **340**, and a power amplifier (PA) **318**. The BBF **312**, the mixer **314**, the DA **316**, the matching network **340**, and the PA **318** may be included in a radio frequency integrated circuit (RFIC). For certain aspects, the PA **318** may be external to the RFIC. In such aspects, the RFIC (and thus the DA **316**) may be coupled to the PA **318** over one or more interconnections, for example, a conductive line or cabling such as a coaxial cable or flex circuit.

[0046] The BBF **312** filters the baseband signals received from the DAC **310**, and the mixer **314** mixes the filtered baseband signals with a transmit local oscillator (LO) signal to convert the baseband signal of interest to a different frequency (e.g., upconvert from baseband to a radio frequency). This frequency-conversion process produces the sum and difference frequencies between the LO frequency and the frequencies of the baseband signal of interest. The sum and difference frequencies are referred to as the "beat frequencies." The beat frequencies are typically in the RF range, such that the signals output by the mixer **314** are typically RF signals, which may be amplified by the DA **316** and/or by the PA **318** before transmission by the antenna(s) **306**. While one mixer **314** is illustrated, several mixers may be used to upconvert the filtered baseband signals to one or more intermediate frequencies and to thereafter upconvert the intermediate frequency (IF) signals to a frequency for transmission.

[0047] The matching network **340** may transfer (amplified) signals (e.g., RF signals) output from the DA **316** to the PA **318**, which may amplify the signals (e.g., RF signals) before transmission by the antenna(s) **306**. The matching network **340** may be used for impedance matching between the DA **316** and PA **318**. For example, the matching network **340** may be used to match an output impedance of the DA **316** to an input impedance of the PA **318**. In some cases, impedance matching can be used to reduce losses and increase the power transferred from the DA **316** and PA **318** to the antenna(s) **306**. In certain aspects, the matching network **340** may be implemented with a transformer, a first coil selectively coupled, inductively, to the transformer, and one or more inductive paths coupled between an input node and an output node of the transformer. Each inductive path may include a respective second coil that is selectively coupled, inductively, to the transformer.

[0048] The RX path **304** may include a low noise amplifier (LNA) **324**, a mixer **326**, and a baseband filter (BBF) **328**. The LNA **324**, the mixer **326**, and the BBF **328** may be included in one or more RFICs, which may or may not be the same RFIC that includes the TX path components. RF signals received via the antenna(s) **306** may be amplified by the LNA **324**, and the mixer **326** mixes the amplified RF signals with a receive local oscillator (LO) signal to convert the RF signal of interest to a different baseband frequency (e.g., downconvert). The baseband signals output by the mixer **326** may be filtered by the BBF **328** before being converted by an analog-to-digital converter (ADC) **330** to digital I and/or Q signals for digital signal processing.

[0049] Certain transceivers may employ frequency synthesizers with a variable-frequency oscillator (e.g., a voltage-controlled oscillator (VCO) or a digitally controlled oscillator (DCO)) to generate a stable, tunable LO with a particular tuning range. Thus, the transmit LO may be produced by a TX frequency synthesizer **320**, which may be buffered or amplified by amplifier **322** before being mixed with the baseband signals in the mixer **314**. Similarly, the receive LO may be produced by an RX frequency synthesizer **332**, which may be buffered or amplified by amplifier **334** before being mixed with the RF signals in the mixer **326**. For certain aspects, a single frequency synthesizer may be used for both the TX path **302** and the RX path **304**. In certain aspects, the TX frequency synthesizer **320** and/or RX frequency synthesizer **332** may include a frequency multiplier, such as a frequency doubler, that is driven by an oscillator (e.g., a VCO) in the frequency synthesizer.

[0050] A controller **336** (e.g., controller/processor **280** in FIG. **2**) may direct the operation of the RF transceiver circuit **300**A, such as transmitting signals via the TX path **302** and/or receiving signals via the RX path **304**. The controller **336** may be a processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof. A memory **338** (e.g., memory **282** in FIG. **2**) may store data and/or program codes for operating the RF transceiver circuit **300**. The controller **336** and/or the memory **338** may include control logic (e.g., complementary metal-oxide-semiconductor (CMOS) logic).

[0051] While FIGS. **1-3** provide wireless communications as an example application in which certain aspects of the present disclosure may be implemented to facilitate understanding, certain aspects described herein may be used for power amplifier circuits in any of various other suitable systems (e.g., an audio system or other electronic system).

Example Matching Network for a Power Amplifier

[0052] As noted above, the PA **318** may be used to amplify signals (e.g., RF signals) before transmission by the antenna **306**. One issue with some PA circuits is that these circuits can suffer breakdown and/or failure of circuit components when operated in certain conditions. In particular, large voltage swings across a power device (e.g., transistor(s)) of such a PA can degrade the ruggedness of the PA, causing part failure.

[0053] In certain cases, impedance matching can be used to improve the ruggedness of a PA. Implementing impedance matching can involve incorporating a matching network prior to the input of the PA to reduce losses and increase the power transferred to the PA. As illustrated in FIG. **3**, for example, the matching network **340** is coupled between the DA **316** and PA **318**. The DA **316** may be representative of a first stage of an amplifier circuit (and may be referred to herein as a "pre-PA"), and the PA **318** may be representative of a second stage of the amplifier circuit.

[0054] FIG. **4** illustrates an example amplifier circuit **400** (or amplification circuit) that may be included within an RF transceiver, such as RF transceiver circuit **300**. The amplifier circuit **400** is one reference example of a multi-stage amplifier circuit with an interstage matching network. The amplifier circuit **400** includes, without limitation, an amplifier **416**, a matching network **440**, and an amplifier **418**. The amplifier **416** may be used to implement the DA **316** of FIG. **3**, the matching network **440** may be used to implement the matching network **340** of FIG. **3**, and the amplifier **418** may be used to implement the PA **318** of FIG. **3**.

[0055] As shown, the matching network **440** is coupled between the amplifier **416** (e.g., first stage) and the amplifier **418** (e.g., second stage). The matching network **440** is configured to transfer a signal from the amplifier **416** to the amplifier **418**. For example, the matching network **440** may (a) receive, via a first input node (IN_1) and a second input node (IN_2) in the matching network **440**, a dual-ended output from the amplifier **418**, and (b) provide, via a first output node (OUT_1) and a second output node (OUT_2) in the matching network **440**, the dual-ended output to the amplifier **418**.

[0056] The matching network **440** includes a transformer **402**, a switchable inductor ring **404**, and tunable capacitive element C_{in}. The transformer **402** includes a primary winding (L_{pri}) and a secondary winding (L_{sec}). A center tap of the primary winding (L_{pri}) may be coupled to a supply voltage (e.g., a reference potential labeled "V_{DD}"). A center tap of the secondary winding (L_{sec}) may be coupled to a bias voltage (e.g., a reference potential labeled "V_{bias}"). The primary winding (L_{pri}) is coupled between the first input node (IN_1) and the second input node (IN_2) of the transformer **402**. The secondary winding (L_{sec}) is coupled between the first output node (OUT_1) and the second output node (OUT_2) of the transformer **402**. The tunable capacitive element C_{in} is coupled in parallel with the primary winding (L_{pri}) of the transformer **402**. In certain implementations, the amplifier **416** and/or the amplifier **418** may include an array of amplifier cells (or slices), where each amplifier cell (or slice) is implemented with a respective transistor. In these implementations, the transformer **402** may tap to the center of the respective array.

[0057] The switchable inductor ring **404** includes a coil L_1 and a switch **410**. In some cases, at least a portion of the switchable inductor ring **404** (including the coil L_1 and/or switch **410**) may be disposed around the primary winding (L_{pri}) and secondary winding (L_{sec}) of the transformer **402**. The coil L_1 may be selectively coupled, inductively, to the transformer **402** via the switch **410**. For example, the switch **410** may inductively couple the coil L_1 to the transformer **402** when the switch **410** is in a closed state, and may inductively decouple the coil L_1 from the transformer **402** when the switch **410** is in an open state. In certain cases, the switch **410** may be implemented by a transmission gate or any of other various suitable components, such as a field-effect transistor (FET) (with a p-type metal-oxide-semiconductor (PMOS) implementation or n-type metal-oxide-semiconductor (NMOS) implementation), negative-positive-negative (NPN) transistor, or positive-negative-positive (PNP) transistor, as illustrative, non-limiting examples. In certain cases, the switch **410** may be configured to be in a closed state or open state in response to receiving a control signal from a controller, such as controller **336**.

[0058] Note that the amplifier circuit **400** depicted in FIG. **4** is provided as a reference example of an amplifier circuit that may be included in an RF transceiver. For example, while the amplifier circuit **400** depicts the matching network **440** with a switchable inductor ring **404**, aspects of the present disclosure may allow for implementing the matching network without a switchable inductor ring. For instance, such a matching network may be implemented with a transformer (e.g., similar to transformer **402**) and a tunable capacitive element (e.g., similar to tunable capacitive element C_{in}).

[0059] In certain cases, the inductance of the transformer **402** may be reduced when the coil L_1 is inductively coupled to the transformer **402** (e.g., due to the switch **410** being in a closed state). For instance, in some cases, the inductive coupling between the coil L_1 and the transformer **402** may reduce the transformer inductance by approximately 50%, resonant with C_{in} at higher frequencies. However, one potential drawback to using a matching network **440** with the switchable inductor ring **404** is that it can impact performance of the amplifier **418** in certain frequency bands (e.g., certain frequency bands associated with the 802.11be wireless communication standard, commonly referred to as "Wi-Fi 7").

[0060] For instance, while the inductive coupling between the coil L_1 and the transformer **402** (e.g., when switch **410** is in the closed state) may reduce the transformer inductance, this inductive coupling may also significantly reduce the Q factor (e.g., from 10 to 6). This decrease in the quality factor can cause gain variation and/or degradation in the amplifier **418**.

[0061] Additionally, the inductive coupling between the coil L_1 and the transformer **402** may impact wideband operation of the amplifier **418**. For example, there may be a reduction in the highest gain peak frequency that can be achieved by the amplifier **418**. In one reference example, the highest gain peak frequency may be approximately 6.7 gigahertz (GHz), which is lower than the 6 GHz band edge (e.g., 7.25 GHZ) associated with 802.11be. Additionally, in some cases, there may be a large gain variation across frequency (e.g., greater than 4 dB gain variation) when the coil L_1 is inductively coupled to the transformer **402**.

[0062] Additionally, in some cases, there may be a large degradation in the saturated output power (Psat) of the amplifier **418** when the coil L_1 is inductively coupled to the transformer **402**. The Psat degradation may be more pronounced at higher frequencies (e.g., 7.25 GHZ).

[0063] Additionally, in cases where the amplifier **418** is implemented with an array of amplifier cells, the inductive coupling between the coil L_1 and the transformer **402** may result in a difference in the array impedance seen by one or more of the amplifier cells. For example, because the transformer **402** may tap to the center of the array, the amplifier cells (or slices) located away from the center (e.g., at the edge of the array) may be presented with a different input impedance (Zin) due to additional inductance from the long array input trace (e.g., metal routing). The difference in the input impedance seen by the amplifier cells may increase as the frequency increases, limiting the tuning range of the matching network **440**.

[0064] To address the aforementioned technical challenges, certain aspects herein describe a matching network that is implemented, in part, with a transformer, a switchable inductor ring, and one or more independently controlled switchable inductor paths. Each switchable inductor path may include a respective coil that is selectively coupled, inductively, to the transformer. In certain aspects, the switchable inductor paths may be added at the edge(s) of the amplifier array input trace to enhance wideband operations.

[0065] Implementing the matching network described herein with the independently controlled switchable inductor paths may improve (e.g., extend) the peak gain frequency, flatten (e.g., decrease) the gain variation across frequency, increase the saturated output power (Psat) at higher frequencies, and improve the Q factor (without a significant degradation in Psat), as illustrative, non-limiting examples. For example, including the switchable inductor paths in the matching network can reduce the impact on the transformer inductance by the switchable inductor ring, which in turn, can reduce the degradation in the Q factor. In addition, implementing the matching network with independently controlled switchable inductor paths may allow for distributed impedance matching. The distributed impedance matching may provide a direct path to center PA array cells and edge PA array cells, mitigating the unwanted inductance edge PA array cells see due to the long array input trace.

[0066] FIG. **5** illustrates an example amplifier circuit **500** (or amplification circuit) that may be included within an RF transceiver, such as RF transceiver circuit **300**, according to certain aspects of the present disclosure. The amplifier circuit **500** is one reference example of a multi-stage amplifier circuit with an interstage matching network. Compared to amplifier circuit **400** of FIG. **4**, the amplifier circuit **500** includes a matching network **540**, which may be used to implement the matching network **340** of FIG. **3**. Compared to the matching network **440** of FIG. **4**, the matching network **540** also includes one or more one or more switchable inductor paths **510-1** to **510**-K and one or more switchable inductor paths **520-1** to **520**-N (e.g., in addition to the transformer **420** and the switchable inductor ring **404**).

[0067] Each switchable inductor path **510** may be coupled between the first input node (IN_1) and the first output node (OUT_1) of the transformer **402**. As shown, each inductor path **510** may include a coil L_2, a capacitive element C_1, a capacitive element C_2, and a switch **502**. The capacitive element C_1 may be coupled between the coil L_2 and the switch **502**, and the capacitive element C_2 may be coupled between the switch **502** and the first output node (OUT_1) of the transformer **402**. One or more of the switchable inductor paths **510** may be located external to at least one of the primary winding (L_{pri}) or the secondary winding (L_{sec}) of the transformer **402**. Additionally, in some cases, one or more of the switchable inductor paths **510** may be located external to the coil L_1 of the switchable inductor ring **404**.

[0068] The respective coil L_2 within each inductor path **510** may be selectively coupled, inductively, to the transformer **402** via the respective switch **502** of the inductor path **510**. For example, the switch **502** may inductively couple the coil L_2 to the transformer **402** when the switch **502** is in a closed state, and may inductively decouple the coil L_2 from the transformer **402** when the switch **502** is in an open state. In certain cases, when the switch **502** is in a closed state, there may be a finite mutual coupling between the coil L_2 and at least one of the transformer **402** or the coil L_1 (when switch **410** is in a closed state). In certain cases, the switch **502** may be implemented by a transmission gate or any of other various suitable components, such as a FET (with a PMOS implementation or NMOS implementation), NPN transistor, or PNP transistor, as illustrative, non-limiting examples. In certain cases, the switch **502** may be configured to be in a closed state or open state in response to receiving a control signal from a controller, such as controller **336**. The capacitive elements C_1 and C_2 may allow for cross-biasing the switch **502**, e.g., to prevent the switch **502** from inadvertently changing states. Additionally, in some cases, the capacitive elements C_1 and C_2 may be used as tuning capacitive elements to tune the impedance of the inductor path **510** (e.g., capacitive elements C_1 and C_2 may have small capacitor values to create a series LC tuning element when the switch **502** is in a closed state).

[0069] Each switchable inductor path **520** may be coupled between the second input node (IN_2) and the second output node (OUT_2) of the transformer **402**. As shown, each inductor path **520** may include a coil L_3, a capacitive element C_3, a capacitive element C_4, and a switch **504**. The capacitive element C_3 may be coupled between the coil L_3 and the switch **504**, and the capacitive element C_4 may be coupled between the switch **504** and the second output node (OUT_2) of the transformer **402**. One or more of the switchable inductor paths **520** may be located external to at least one of the primary winding (L_{pri}) or the secondary winding (L_{sec}) of the transformer **402**. Additionally, in some cases, one or more of the switchable inductor paths **520** may be located external to the coil L_1 of the switchable inductor ring **404**.

[0070] The respective coil L_3 within each inductor path **520** may be selectively coupled, inductively, to the transformer **402** via the respective switch **504** of the inductor path **520**. For example, the switch **504** may inductively couple the coil L_3 to the transformer **402** when the switch **504** is in a closed state, and may inductively decouple the coil L_3 from the transformer **402** when the switch **504** is in an open state. In certain cases, when the switch **504** is in a closed state, there may be a finite mutual coupling between the coil L_3 and at least one of the transformer **402** or the coil L_1 (when switch **410** is in a closed state). In certain cases, the switch **504** may be implemented by a transmission gate or any of other various suitable components, such as a FET (with a PMOS implementation or NMOS implementation), NPN transistor, or PNP transistor, as illustrative, non-limiting examples. In certain cases, the switch **504** may be configured to be in a closed state or open state in response to receiving a control signal from a controller, such as controller **336**. The capacitive elements C_3 and C_4 may allow for cross-biasing the switch **504**, e.g., to prevent the switch **504** from inadvertently changing states. Additionally, in some cases, the capacitive elements C_3 and C_4 may be used as tuning capacitive elements to tune the impedance of the inductor path **520** (e.g., capacitive elements C_3 and C_4 may have small capacitor values to create a series LC tuning element when the switch **504** is in a closed state).

[0071] Note that the number of switchable inductor paths **510** may be the same as, or different than, the number of switchable inductor paths **520**. For example, in some aspects, the number of switchable inductor paths **510** may be based on the number of edge amplifier cells located at or near a first end of the PA array, e.g., with respect to the center of the PA array. Similarly, the number of switchable inductor paths **520** may be based on the number of edge amplifier cells located at or near a second end of the PA array, e.g., with respect to the center of the PA array. In some cases, the matching network **540** may include a respective switchable inductor path **510** for each amplifier cell located at or near the first end of the PA array, and a respective switchable inductor path **510** for each amplifier cell located at or near the second end of the PA array. In one reference example, the switchable inductor path **510-1** may be located near a first edge of the PA array to mitigate unwanted inductance to which one or more of the edge amplifier cells in the PA array may be presented due to a long array input trace. Likewise, the switchable inductor path **520-1** may be located near a second opposite edge of the PA array to mitigate unwanted inductance to which one or more of the edge amplifier cells in the PA array may be presented due to the long array input trace.

[0072] Additionally, note that the amplifier circuit **500** depicted in FIG. **5** is provided as a reference example of an amplifier circuit that may be included in an RF transceiver. For example, while the amplifier circuit **500** depicts the matching network **540** with a switchable inductor ring **404**, aspects of the present disclosure may allow for implementing the matching network without a switchable inductor ring. For instance, such a matching network may be implemented with a transformer (e.g., similar to transformer **402**), a tunable capacitive element (e.g., similar to tunable capacitive element C_{in}), one or more first switchable inductor paths (e.g., similar to switchable inductor paths **510-1** to **510**-K), and one or more second switchable inductor paths (e.g., similar to switchable inductor paths **520-1** to **520**-N).

[0073] In certain aspects, each of the respective switches **502** of the switchable inductor paths **510**, each of the respective switches **504** of the switchable inductor paths **520**, and/or the switch **410** of the switchable inductor ring **404** may be independently controlled, via the controller **336**. For example, each of the switches **502**, **504**, and **410** can be independently controlled to achieve a target amplifier performance.

[0074] FIG. **6** illustrates a graph **600** of different final tuned gains of an amplifier circuit (e.g., PA **318**, amplifier **418**, etc.) for different types of matching networks. In particular, graph **600** depicts (i) the tuned gain **610** of the amplifier circuit for a matching network implemented with a transformer **402** only and (ii) the tuned gain **620** of the amplifier circuit for a matching network implemented with a transformer **402**, switchable inductor ring **404** having switch **410** in a closed state, switchable inductor path(s) **510** having switch(es) **502** in a closed state, and switchable inductor path(s) **520** having switch(es) **504** in a closed state. Note that the tuned gain **610** may be achieved with a matching network implemented with a transformer **402** only (e.g., without a switchable inductor ring **404**, switchable inductor path(s) **510**, and switchable inductor path(s) **520**) or with a matching network implemented with a transformer **402**, switchable inductor ring **404** having switch **410** in an open state, switchable inductor path(s) **510** having switch(es) **502** in an open state, and switchable inductor path(s) **520** having switch(es) **504** in an open state. In certain aspects, the amplifier circuit may have improved performance with a matching network described in various aspects of the present disclosure, such as matching network **540**. As shown in graph **600**, for example, the tuned gain **620** of the amplifier circuit has a single, more defined peak relative to the tuned gain **610** of the amplifier circuit.

Example Operations

[0075] FIG. **7** is a flow diagram of example operations **700** for amplifying a signal for transmission, in accordance with certain aspects of the present disclosure. The operations **700** may be performed, for example, by an amplifier circuit (or amplification circuit) (e.g., amplifier circuit **500**) of a transceiver (e.g., transceiver **232** and/or transceiver **254**).

[0076] The operations **700** may generally involve, at block **702**, transferring a signal from a first amplifier (e.g., amplifier **316** or **416**) to a second amplifier (e.g., amplifier **416** or **418**) via a matching network (e.g., matching network **540**) of the amplifier circuit.

[0077] The matching network may include a transformer (e.g., transformer **402**) including: a first input node (e.g., first input node IN_1) and a second input node (e.g., second input node IN_2) for coupling to a first output and a second output of the first amplifier; a first output node (e.g., first output node OUT_1) and a second output node (e.g., second output node OUT_2) for coupling to a first input and a second input of the second amplifier; a primary winding (e.g., primary winding L_{pri}) coupled between the first input node and the second input node of the transformer; and a secondary winding (e.g., secondary winding L_{sec}) inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer. The matching network may also include a first coil (e.g., coil L_1) selectively coupled, inductively, to the transformer.

[0078] The matching network may further include one or more first inductive paths (e.g., switchable inductor paths **510**) coupled between the first input node and the first output node of the transformer. Each inductive path of the one or more first inductive paths may include a respective second coil (e.g., coil L_2) selectively coupled, inductively, to the transformer.

[0079] In some aspects, the matching network may further include one or more second inductive paths (e.g., switchable inductor paths **520**) coupled between the second input node and the second output node of the transformer. Each inductive path of the one or more second inductive paths may include a respective third coil (e.g., coil L_3) selectively coupled, inductively, to the transformer.

[0080] The operations **700** may also involve, at block **704**, amplifying, via the second amplifier, the signal transferred via the matching network.

[0081] In certain aspects, transferring the signal (at block **702**) may involve at least one of (i) selectively coupling the first coil, inductively, to the transformer or (ii) selectively coupling at least one respective second coil of the one or more inductive paths, inductively, to the transformer. In some aspects, the selectively coupling of the at least one respective second coil, inductively, to the transformer may be independent of the selectively coupling of at least another respective second coil of the one or more inductive paths, inductively, to the transformer. Additionally or alternatively, in some aspects, the selectively coupling of the first coil, inductively, to the transformer may be independent of the selectively coupling of the at least one respective second coil of the one or more inductive paths, inductively, to the transformer.

Example Aspects

[0082] In addition to the various aspects described above, specific combinations of aspects are within the scope of the present disclosure, some of which are detailed below:

[0083] Aspect 1: A matching network for an amplification circuit, comprising: a transformer comprising: a first input node and a second input node for coupling to a first stage of the amplification circuit; a first output node and a second output node for coupling to a second stage of the amplification circuit; a primary winding coupled between the first input node and the second input node of the transformer; and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer; a first coil selectively coupled, inductively, to the transformer; and one or more first inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more first inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer.

[0084] Aspect 2: The matching network of Aspect 1, further comprising one or more second inductive paths coupled between the second input node and the second output node of the transformer, each inductive path of the one or more second inductive paths comprising a respective third coil selectively coupled, inductively, to the transformer.

[0085] Aspect 3: The matching network of Aspect 2, wherein the one or more second inductive paths comprise a plurality of inductive paths, each coupled in parallel between (i) the second input node of the transformer and (ii) the second output node of the transformer.

[0086] Aspect 4: The matching network of any of Aspects 1 to 3, wherein the one or more first inductive paths comprise a plurality of inductive paths, each coupled in parallel between (i) the first input node of the transformer and (ii) the first output node of the transformer.

[0087] Aspect 5: The matching network of any of Aspects 1 to 4, wherein the first coil is disposed around the primary winding and the secondary winding of the transformer.

[0088] Aspect 6: The matching network of any of Aspects 1 to 5, further comprising a switch comprising (i) a first terminal coupled to a first terminal of the first coil and (ii) a second terminal coupled to a second terminal of the first coil, wherein the switch is configured to inductively couple the first coil to the transformer in a closed state and wherein the switch is configured to inductively decouple the first coil from the transformer in an open state.

[0089] Aspect 7: The matching network of any of Aspects 1 to 6, wherein each inductive path of the one or more first inductive paths further comprises a respective switch comprising a first terminal coupled to the respective second coil of the inductive path and a second terminal coupled to the first output node of the transformer, wherein the respective switch is configured to inductively couple the respective second coil to the transformer in a closed state, and wherein the respective switch is configured to inductively decouple the respective second coil from the transformer in an open state.

[0090] Aspect 8: The matching network of Aspect 7, wherein each inductive path of the one or more first inductive paths further comprises: a respective first capacitive element coupled between the respective second coil of the inductive path and the respective switch of the inductive path; and a respective second capacitive element coupled between the respective switch of the inductive path and the first output node of the transformer.

[0091] Aspect 9: The matching network of any of Aspects 1 to 8, wherein the one or more first inductive paths are located external to at least one of the primary winding or the secondary winding of the transformer.

[0092] Aspect 10: The matching network of any of Aspects 1 to 9, wherein the one or more first inductive paths are located external to the first coil.

[0093] Aspect 11: The matching network of any of Aspects 1 to 10, wherein the first coil is disposed around the primary winding and the secondary winding of the transformer.

[0094] Aspect 12: The matching network of any of Aspects 1 to 11, further comprising a tunable capacitive element coupled in parallel with the primary winding of the transformer.

[0095] Aspect 13: A wireless device comprising: a first amplifier; a second amplifier; and a matching network coupled between the first amplifier and the second amplifier, the matching network comprising: a transformer comprising: a first input node and a second input node for coupling to a first output and a second output of the first amplifier; a first output node and a second output node for coupling to a first input and a second input of the second amplifier; a primary winding coupled between the first input node and the second input node of the transformer; and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer; a first coil selectively coupled, inductively, to the transformer; and one or more first inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more first inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer.

[0096] Aspect 14: The wireless device of Aspect 13, wherein the matching network further comprises one or more second inductive paths coupled between the second input node and the second output node of the transformer, each inductive path of the one or more second inductive paths comprising a respective third coil selectively coupled, inductively, to the transformer.

[0097] Aspect 15: The wireless device of any of Aspects 13 to 14, wherein the matching network further comprises a switch comprising (i) a first terminal coupled to a first terminal of the first coil and (ii) a second terminal coupled to a second terminal of the first coil, wherein the switch is configured to inductively couple the first coil to the transformer in a closed state and wherein the switch is configured to inductively decouple the first coil from the transformer in an open state.

[0098] Aspect 16: The wireless device of any of Aspects 13 to 15, wherein each inductive path of the one or more first inductive paths further comprises a respective switch comprising a first terminal coupled to the respective second coil of the inductive path and a second terminal coupled to the first output node of the transformer, wherein the respective switch is configured to inductively couple the respective second coil to the transformer in a closed state, and wherein the respective switch is configured to inductively decouple the respective second coil from the transformer in an open state.

[0099] Aspect 17: A method of wireless communication, comprising: transferring a signal from a first amplifier to a second amplifier via a matching network, wherein the matching network comprises: a transformer comprising: a first input node and a second input node for coupling to a first output and a second output of the first amplifier; a first output node and a second output node for coupling to a first input and a second input of the second amplifier; a primary winding coupled between the first input node and the second input node of the transformer; and a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer; a first coil selectively coupled, inductively, to the transformer; and one or more inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer; and amplifying, via the second amplifier, the signal transferred via the matching network.

[0100] Aspect 18: The method of Aspect 17, wherein transferring the signal from the first amplifier to the second amplifier via the matching network comprises at least one of: selectively coupling the first coil, inductively, to the transformer; or selectively coupling at least one respective second coil of the one or more inductive paths, inductively, to the transformer.

[0101] Aspect 19: The method of Aspect 18, wherein the selectively coupling of the at least one respective second coil, inductively, to the transformer is independent of the selectively coupling of at least another respective second coil of the one or more inductive paths, inductively, to the transformer.

[0102] Aspect 20: The method of any of Aspects 18 to 19, wherein the selectively coupling of the first coil, inductively, to the transformer is independent of the selectively coupling of the at least one respective second coil of the one or more inductive paths, inductively, to the transformer.

[0103] The above description provides examples, and is not limiting of the scope, applicability, or examples set forth in the claims. Changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects.

[0104] The various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application-specific integrated circuit (ASIC), or processor. Generally, where there are operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components. For example, means for transferring a signal from a first amplifier to a second amplifier may include a matching network, such as the matching network **340** of FIG. **3** or the matching network **540** of FIG. **5**. Means for selectively coupling may include a switch, such as the switch **410** of FIG. **4**, the switch **502** of FIG. **5**, or the switch **504** of FIG. **5**. Means for controlling a switch may include a controller (or processor), such as the controller **336** of FIG. **3**. Means for amplifying the signal transferred via the matching network may include an amplifier, such as the PA **318** of FIG. **3** or the amplifier **418** of FIG. **4** or **5**.

[0105] As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

[0106] As used herein, "a processor," "at least one processor," or "one or more processors" generally refers to a single processor configured to perform one or multiple operations or multiple processors configured to collectively perform one or more operations. In the case of multiple processors, performance of the one or more operations could be divided amongst different processors, though one processor may perform multiple operations, and multiple processors could collectively perform a single operation. Similarly, "a memory," "at least one memory," or "one or more memories" generally refers to a single memory configured to store data and/or instructions or multiple memories configured to collectively store data and/or instructions.

[0107] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0108] It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes, and variations may be made in the arrangement, operation, and details of the methods and apparatus described above without departing from the scope of the claims.

1. A matching network for an amplification circuit, comprising:
- a transformer comprising:
  - a first input node and a second input node for coupling to a first stage of the amplification circuit;
  - a first output node and a second output node for coupling to a second stage of the amplification circuit;
  - a primary winding coupled between the first input node and the second input node of the transformer; and
  - a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer;
- a first coil selectively coupled, inductively, to the transformer; and
- one or more first inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more first inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer.

2. The matching network of claim **1**, further comprising one or more second inductive paths coupled between the second input node and the second output node of the transformer, each inductive path of the one or more second inductive paths comprising a respective third coil selectively coupled, inductively, to the transformer.

3. The matching network of claim **2**, wherein the one or more second inductive paths comprise a plurality of inductive paths, each coupled in parallel between (i) the second input node of the transformer and (ii) the second output node of the transformer.

4. The matching network of claim **1**, wherein the one or more first inductive paths comprise a plurality of inductive paths, each coupled in parallel between (i) the first input node of the transformer and (ii) the first output node of the transformer.

5. The matching network of claim **1**, wherein the first coil is disposed around the primary winding and the secondary winding of the transformer.

6. The matching network of claim **1**, further comprising a switch comprising (i) a first terminal coupled to a first terminal of the first coil and (ii) a second terminal coupled to a second terminal of the first coil, wherein the switch is configured to inductively couple the first coil to the transformer in a closed state and wherein the switch is configured to inductively decouple the first coil from the transformer in an open state.

7. The matching network of claim **1**, wherein each inductive path of the one or more first inductive paths further comprises a respective switch comprising a first terminal coupled to the respective second coil of the inductive path and a second terminal coupled to the first output node of the transformer, wherein the respective switch is configured to inductively couple the respective second coil to the transformer in a closed state, and wherein the respective switch is configured to inductively decouple the respective second coil from the transformer in an open state.

8. The matching network of claim **7**, wherein each inductive path of the one or more first inductive paths further comprises:

a respective first capacitive element coupled between the respective second coil of the inductive path and the respective switch of the inductive path; and

a respective second capacitive element coupled between the respective switch of the inductive path and the first output node of the transformer.

9. The matching network of claim **1**, wherein the one or more first inductive paths are located external to at least one of the primary winding or the secondary winding of the transformer.

10. The matching network of claim **1**, wherein the one or more first inductive paths are located external to the first coil.

11. The matching network of claim **10**, wherein the first coil is disposed around the primary winding and the secondary winding of the transformer.

12. The matching network of claim **1**, further comprising a tunable capacitive element coupled in parallel with the primary winding of the transformer.

13. A wireless device comprising:

a first amplifier;

a second amplifier; and

a matching network coupled between the first amplifier and the second amplifier, the matching network comprising:

a transformer comprising:

a first input node and a second input node for coupling to a first output and a second output of the first amplifier;

a first output node and a second output node for coupling to a first input and a second input of the second amplifier;

a primary winding coupled between the first input node and the second input node of the transformer; and

a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer;

a first coil selectively coupled, inductively, to the transformer; and

one or more first inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more first inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer.

14. The wireless device of claim **13**, wherein the matching network further comprises one or more second inductive paths coupled between the second input node and the second output node of the transformer, each inductive path of the one or more second inductive paths comprising a respective third coil selectively coupled, inductively, to the transformer.

15. The wireless device of claim **13**, wherein the matching network further comprises a switch comprising (i) a first terminal coupled to a first terminal of the first coil and (ii) a second terminal coupled to a second terminal of the first coil, wherein the switch is configured to inductively couple the first coil to the transformer in a closed state and wherein the switch is configured to inductively decouple the first coil from the transformer in an open state.

16. The wireless device of claim **13**, wherein each inductive path of the one or more first inductive paths further comprises a respective switch comprising a first terminal coupled to the respective second coil of the inductive path and a second terminal coupled to the first output node of the transformer, wherein the respective switch is configured to inductively couple the respective second coil to the transformer in a closed state, and wherein the respective switch is configured to inductively decouple the respective second coil from the transformer in an open state.

17. A method of wireless communication, comprising:

transferring a signal from a first amplifier to a second amplifier via a matching network, wherein the matching network comprises:

a transformer comprising:

a first input node and a second input node for coupling to a first output and a second output of the first amplifier;

a first output node and a second output node for coupling to a first input and a second input of the second amplifier;

a primary winding coupled between the first input node and the second input node of the transformer; and

a secondary winding inductively coupled to the primary winding and coupled between the first output node and the second output node of the transformer;

a first coil selectively coupled, inductively, to the transformer; and

one or more inductive paths coupled between the first input node and the first output node of the transformer, each inductive path of the one or more inductive paths comprising a respective second coil selectively coupled, inductively, to the transformer; and

amplifying, via the second amplifier, the signal transferred via the matching network.

18. The method of claim **17**, wherein transferring the signal from the first amplifier to the second amplifier via the matching network comprises at least one of:

selectively coupling the first coil, inductively, to the transformer; or

selectively coupling at least one respective second coil of the one or more inductive paths, inductively, to the transformer.

19. The method of claim **18**, wherein the selectively coupling of the at least one respective second coil, inductively, to the transformer is independent of the selectively coupling of at least another respective second coil of the one or more inductive paths, inductively, to the transformer.

20. The method of claim **18**, wherein the selectively coupling of the first coil, inductively, to the transformer is independent of the selectively coupling of the at least one respective second coil of the one or more inductive paths, inductively, to the transformer.

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